

Lecture 33 One-way ANOVA Examples

BIO210 Biostatistics

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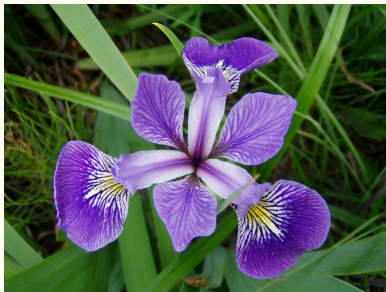
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The Iris Flower Dataset

- Introduced by Ronald Fisher in his 1936 paper: **The use of multiple measurements in taxonomic problems.**
- Extensively used in the machine learning community for testing classification methods. https://en.wikipedia.org/wiki/Iris_flower_data_set



Setosa

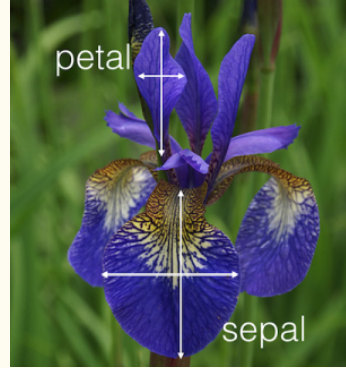
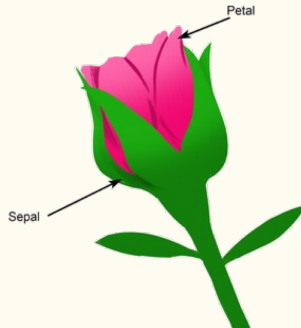
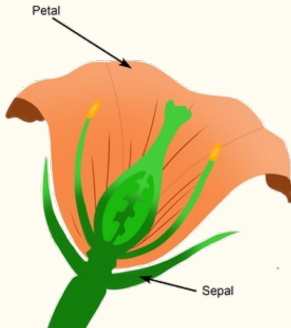


Versicolor



Virginica

The Iris Flower Dataset



The Iris Flower Dataset

THE USE OF MULTIPLE MEASUREMENTS IN TAXONOMIC PROBLEMS

By R. A. FISHER, Sc.D., F.R.S.

I. DISCRIMINANT FUNCTIONS

When two or more populations have been measured in several characters, $x_1, \dots, x_p$, special interest attaches to certain linear functions of the measurements by which the populations are best discriminated. At the author's suggestion use has already been made of this fact in craniometry (a) by Mr E. S. Martin, who has applied the principle to the sex differences in measurements of the mandible, and (b) by Miss Mildred Barnard, who showed how to obtain from a series of dated series the particular compound of cranial measurements showing most distinctly a progressive or secular trend. In the present paper the application of the same principle will be illustrated on a taxonomic problem; some questions connected with the precision of the processes employed will also be discussed.

II. ARITHMETICAL PROCEDURE

Table I shows measurements of the flowers of fifty plants each of the two species *Iris setosa* and *I. versicolor*, found growing together in the same colony and measured by Dr E. Anderson, to whom I am indebted for the use of the data. Four flower measurements are given. We shall first consider the question: What linear function of the four measurements

$$X = \lambda_1 x_1 + \lambda_2 x_2 + \lambda_3 x_3 + \lambda_4 x_4$$

will maximize the ratio of the difference between the specific means to the standard deviations within species? The observed means and their differences are shown in Table II. We may represent the differences by d_p , where $p = 1, 2, 3, 4$ for the four measurements.

The sums of squares and products of deviations from the specific means are shown in Table III. Since fifty plants of each species were used these sums contain 98 degrees of freedom. We may represent these sums of squares or products by S_{pq} , where p and q take independently the values 1, 2, 3 and 4.

Then for any linear function, X , of the measurements, as defined above, the difference between the means of X in the two species is

$$D = \lambda_1 d_1 + \lambda_2 d_2 + \lambda_3 d_3 + \lambda_4 d_4,$$

while the variance of X within species is proportional to

$$S = \sum_{p=1}^4 \sum_{q=1}^4 \lambda_p \lambda_q S_{pq}.$$

The particular linear function which best discriminates the two species will be one for

180 MULTIPLE MEASUREMENTS IN TAXONOMIC PROBLEMS

Table I

<i>Iris setosa</i>				<i>Iris versicolor</i>				<i>Iris virginica</i>			
Sepal length	Sepal width	Petal length	Petal width	Sepal length	Sepal width	Petal length	Petal width	Sepal length	Sepal width	Petal length	Petal width
5.1	3.5	1.4	0.2	7.0	3.2	4.7	1.4	6.3	3.3	6.0	2.3
4.9	3.0	1.4	0.2	6.4	3.2	4.5	1.5	5.8	2.7	5.1	1.9
4.7	3.2	1.3	0.2	6.9	3.1	4.9	1.5	7.1	3.0	5.9	2.1
4.0	3.1	1.5	0.2	5.5	2.3	4.0	1.3	6.3	2.9	5.6	1.8
5.0	3.6	1.4	0.2	6.3	2.8	4.6	1.5	6.5	3.0	5.8	2.2
5.4	3.9	1.7	0.4	5.7	2.8	4.0	1.3	7.6	3.0	6.6	2.1
4.6	3.4	1.4	0.3	6.3	3.0	5.7	1.6	4.9	2.5	4.5	1.7
5.0	3.4	1.3	0.2	4.9	2.4	3.3	1.0	7.3	2.9	6.3	1.8
4.4	2.9	1.4	0.2	6.8	2.0	4.6	1.3	6.7	2.5	5.8	1.9
4.9	3.1	1.3	0.1	5.2	2.7	3.9	1.4	7.2	2.6	6.1	2.0
5.4	3.7	1.5	0.2	5.0	2.0	3.8	1.6	6.8	2.9	5.1	2.0
4.8	3.4	1.6	0.2	5.9	3.0	4.2	1.5	6.4	2.7	5.3	1.9
4.6	3.0	1.4	0.1	6.0	2.2	4.0	1.6	6.8	3.0	5.5	2.1
4.3	3.0	1.1	0.1	6.1	2.9	4.7	1.4	2.7	2.5	5.0	2.0
4.6	4.0	1.2	0.2	5.6	2.9	3.6	1.3	5.4	2.6	5.1	2.4
5.7	4.4	1.3	0.4	6.7	3.1	6.4	1.4	6.4	2.2	5.3	2.3
5.4	3.9	1.3	0.4	5.6	2.0	4.3	1.3	6.3	2.0	5.1	2.4
5.1	3.5	1.4	0.3	5.8	2.7	4.1	1.0	7.7	3.0	6.7	2.2
5.7	5.8	1.7	0.3	6.2	3.2	6.8	1.5	7.7	2.6	6.6	2.3
5.1	3.8	1.3	0.3	5.9	2.9	3.9	1.1	6.0	2.2	5.0	1.5
5.4	3.4	1.7	0.2	5.0	3.2	4.8	1.8	6.9	3.2	5.7	2.3
5.1	3.7	1.5	0.4	6.1	2.8	4.0	1.3	5.6	2.6	4.9	2.0
4.8	2.6	1.0	0.2	4.5	2.6	4.9	1.5	7.7	2.6	6.7	2.6
5.1	3.3	1.7	0.3	6.1	2.8	4.7	1.2	6.3	2.7	4.9	1.4
4.8	2.4	1.9	0.2	6.4	2.9	4.3	1.3	6.7	3.0	5.7	2.1
4.8	2.9	1.6	0.2	5.6	2.0	4.4	1.4	7.2	3.2	6.0	1.8
5.0	2.4	1.6	0.4	5.8	2.8	4.8	1.4	6.2	2.6	4.6	1.8
5.2	2.6	1.6	0.2	5.6	2.9	4.0	1.7	6.1	3.0	4.9	1.8
5.2	2.4	1.4	0.2	6.0	2.9	4.3	1.5	6.4	2.6	5.6	2.1
4.7	3.2	1.6	0.2	5.7	2.6	2.5	1.0	7.2	3.0	5.8	1.4
4.8	3.1	1.6	0.2	5.5	2.4	3.9	1.1	7.4	2.8	6.1	1.9
5.4	3.4	1.5	0.4	5.6	2.4	3.7	1.0	7.9	3.0	6.4	2.0
4.4	3.1	1.5	0.3	5.8	2.7	3.9	1.2	6.4	2.6	5.2	2.2
5.3	4.2	1.4	0.2	6.6	2.7	5.1	1.6	6.3	2.6	5.1	1.5
4.9	3.1	1.5	0.3	5.5	2.6	4.5	1.5	6.1	2.6	5.4	2.1
5.0	3.2	1.2	0.2	6.9	2.4	4.0	1.6	7.7	3.0	6.1	2.3
5.3	3.0	1.9	0.2	6.7	3.1	4.7	1.5	6.8	3.4	6.0	2.4
4.9	3.9	1.4	0.1	6.3	2.3	4.4	1.3	6.4	3.1	5.5	1.4
4.4	3.0	1.2	0.2	5.6	2.0	4.1	1.3	6.0	2.4	4.8	1.6
5.1	3.4	1.5	0.2	5.8	2.3	4.0	1.3	6.9	3.1	5.4	2.1
5.0	3.0	1.3	0.3	5.5	2.8	4.4	1.2	6.7	3.1	5.4	2.4
4.5	2.3	1.3	0.3	6.1	2.0	4.6	1.4	6.0	3.1	5.1	2.3
4.6	3.5	1.3	0.2	5.4	2.8	4.0	1.2	5.8	2.7	5.1	1.9
5.0	3.0	1.6	0.3	5.6	2.9	3.3	1.0	6.2	3.0	5.9	2.0
5.1	3.6	1.9	0.4	5.4	2.7	4.2	1.3	6.7	3.2	5.7	2.5
4.6	3.0	1.4	0.3	5.7	3.0	4.2	1.2	6.7	3.2	5.7	2.3
5.1	3.6	1.6	0.2	5.7	2.9	4.2	1.3	6.5	2.5	5.9	1.9
4.6	3.0	1.4	0.3	6.2	2.9	4.3	1.3	6.5	2.9	5.8	2.3
5.3	3.7	1.3	0.3	6.1	2.5	3.0	1.1	6.2	3.4	5.4	2.3
5.0	3.3	1.4	0.3	5.7	2.9	4.1	1.2	5.9	3.0	5.1	1.8

The Iris Flower Dataset - Formatting

Typical data input format ($m \times n$ matrix):

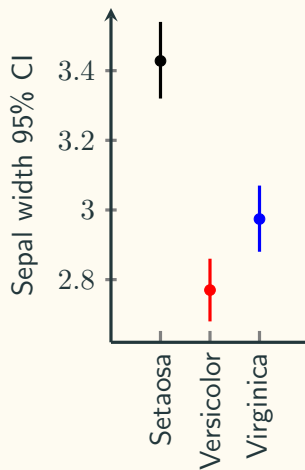
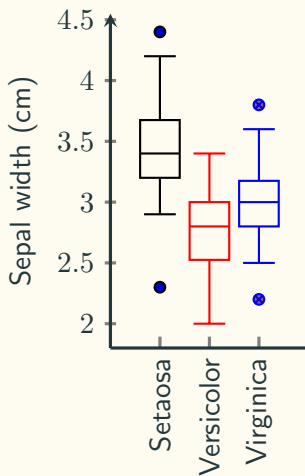
n features						
m observations	1	3	<i>d</i>	<i>k</i>	7	<i>a</i> ...
	8	2	<i>c</i>	8	1	<i>c</i> ...
	7	4	<i>e</i>	<i>x</i>	1	<i>d</i> ...
	9	6	<i>z</i>	<i>y</i>	5	<i>e</i> ...
	5	8	<i>x</i>	<i>z</i>	8	<i>f</i> ...
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

observations: subjects of interest, ~~samples~~
of interest;

features: characteristics describing the observation and they vary among observations.

sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
5.1	3.5	1.4	0.2	setosa
4.9	3.0	1.4	0.2	setosa
4.7	3.2	1.3	0.2	setosa
7.0	3.2	4.7	1.4	versicolor
6.4	3.2	4.5	1.5	versicolor
6.9	3.1	4.9	1.5	versicolor
6.3	3.3	6.0	2.5	virginica
5.8	2.7	5.1	1.9	virginica
7.1	3.0	5.9	2.1	virginica

The Iris Flower Dataset - Plotting



Performing ANOVA Using Statistical Software

- Software choices

- R
- Python
- SAS
- Stata
- SPSS
- Minitab



Fisher's Least Significant Difference (LSD)

When doing *post hoc* pairwise *t*-tests, use the following **test statistic (equal variance) for all comparisons**:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\text{MSW} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}, \text{ where } \nu = n - k$$

Note the difference between s_p^2 and MSW

One-way/factor ANOVA

- **One-way/factor ANOVA:** samples can be distinguished by one factor:
 - Brands of tyres
 - Species
 - *etc.*
- **Two-way/factor ANOVA:** samples can be distinguished by two factors:
 - Brands of tyres + colours
 - Species + location
 - *etc.*